

Changing velocity

Acceleration is a measure of the rate of change in velocity—how long it takes for an object to increase (or decrease) from one velocity to another. Acceleration is calculated by subtracting the starting velocity (V_1) from the final velocity (V_2) to obtain the change in velocity. This figure is then divided by the time that has passed.

rate of change of velocity (meters per second per second)

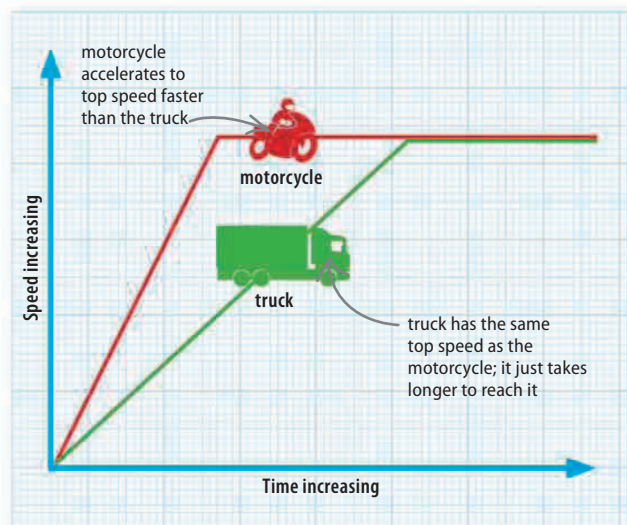
total change in velocity (meters per second)

time in which this change happened (seconds)

$$\text{Acceleration} = \frac{V_2 - V_1}{\text{Time}}$$

► Motorcycle versus truck

This graph shows the greater acceleration of the motorcycle compared to the truck, before they both reach the same cruising speed.



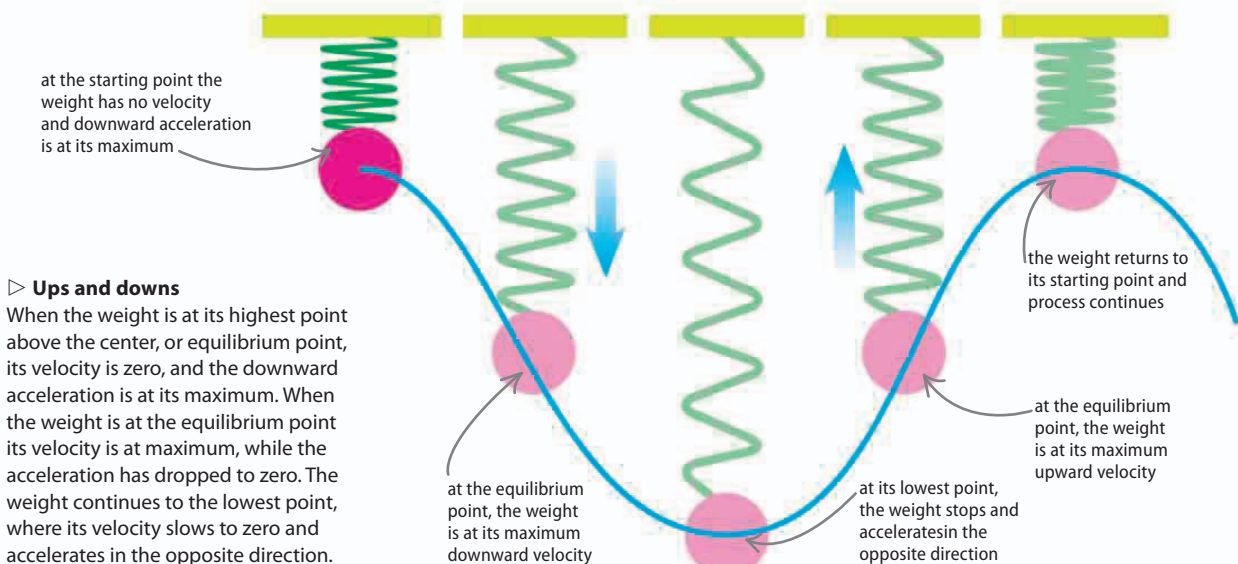
Oscillation

An oscillation is a regular movement about a central point. Whether it is a pendulum swinging from side to side, a weight bouncing on the end of a spring, or the molecules vibrating inside a solid, the motion results from regular accelerations and decelerations. In turn, these produce an average velocity of zero because the object ends up coming back to the same central point. This phenomenon is caused by two opposing forces that accelerate the object to the center, but the resulting velocity moves the same distance in the opposite direction.

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Timekeeping

Oscillations repeat at a constant rate. The time it takes for the oscillator to complete a full cycle is called the period. A pendulum oscillates with a fixed period, and the long pendulums in grandfather clocks have a period of two seconds. Each swing turns the cogs just enough to keep the hands moving at the right rate.



► Ups and downs

When the weight is at its highest point above the center, or equilibrium point, its velocity is zero, and the downward acceleration is at its maximum. When the weight is at the equilibrium point its velocity is at maximum, while the acceleration has dropped to zero. The weight continues to the lowest point, where its velocity slows to zero and accelerates in the opposite direction.